

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1882

Flambeau Hydroelectric Station 100 kV
August 18, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1882. The Interconnection Customer proposes a 55.9 MW natural gas combined cycle facility interconnecting at the Flambeau Hydroelectric Station 100 kV substation in the Illinois area. The request was received on September 18, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 671 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,243,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Sweatt - Arch Smith 100kV Line 1	Arch Smith - Sweatt 100kV L	128.7	241	5.8
Sweatt - Arch Smith 100kV Circuit 2	Sweatt - Arch Smith 100kV	128.7	241	5.8
East Frankfort - Liggett 100kV	Flambeau Hydroelectric Station 100/1kV Transformer Bank	104.7	149	12.8
Davis Crek - Flambeau Hydroelectric Sta 100kV	Flambeau Hydroelectric Sta 100 kV/1 kV TX Unit 1	103.5	285	33.3
University Of Colorado - Flambeau Hydroelectric Station 230kV	Flambeau Hydroelectric Station - Pueblo Xtra 230kV	101.2	442	31.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sweatt - Arch Smith 100kV Line 1	Rebuild line with higher-capacity conductor	\$2,221,000	1% (\$25,000)
Sweatt - Arch Smith 100kV Circuit 2	Rebuild line with higher-capacity conductor	\$1,713,000	1% (\$20,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
East Frankfort - Liggett 100kV	Reconductor line	\$1,888,000	4% (\$79,000)
Davis Crk - Flambeau Hydroelectric Sta 100kV	Reconductor line	\$420,000	5% (\$22,000)
University Of Colorado - Flambeau Hydroelectric Station 230kV	Reconductor line	\$31,036,000	4% (\$1,097,000)
	Total allocated to project		\$1,243,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.